# Iodimetric Titration of Vitamin C ("Back Titration")

Chem 151 Lab

# Equation and Logistics

Vitamin C

$$C_6H_8O_6 + I_3^- \rightarrow C_6H_6O_6 + 2\,H^+ + 3\,I^-$$

Ascorbic acid

LIMITING REAGENT

IN EXCESS
(we make this)

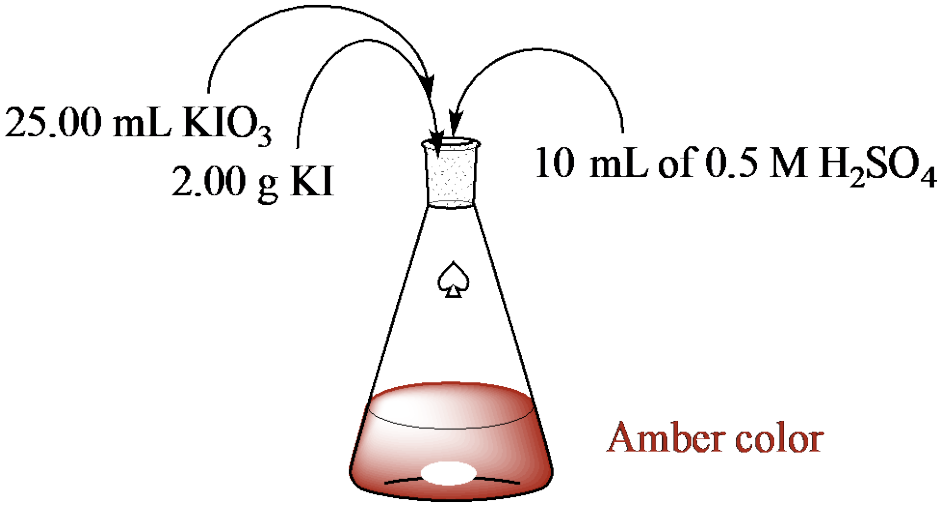
Titrate to determine amount in excess

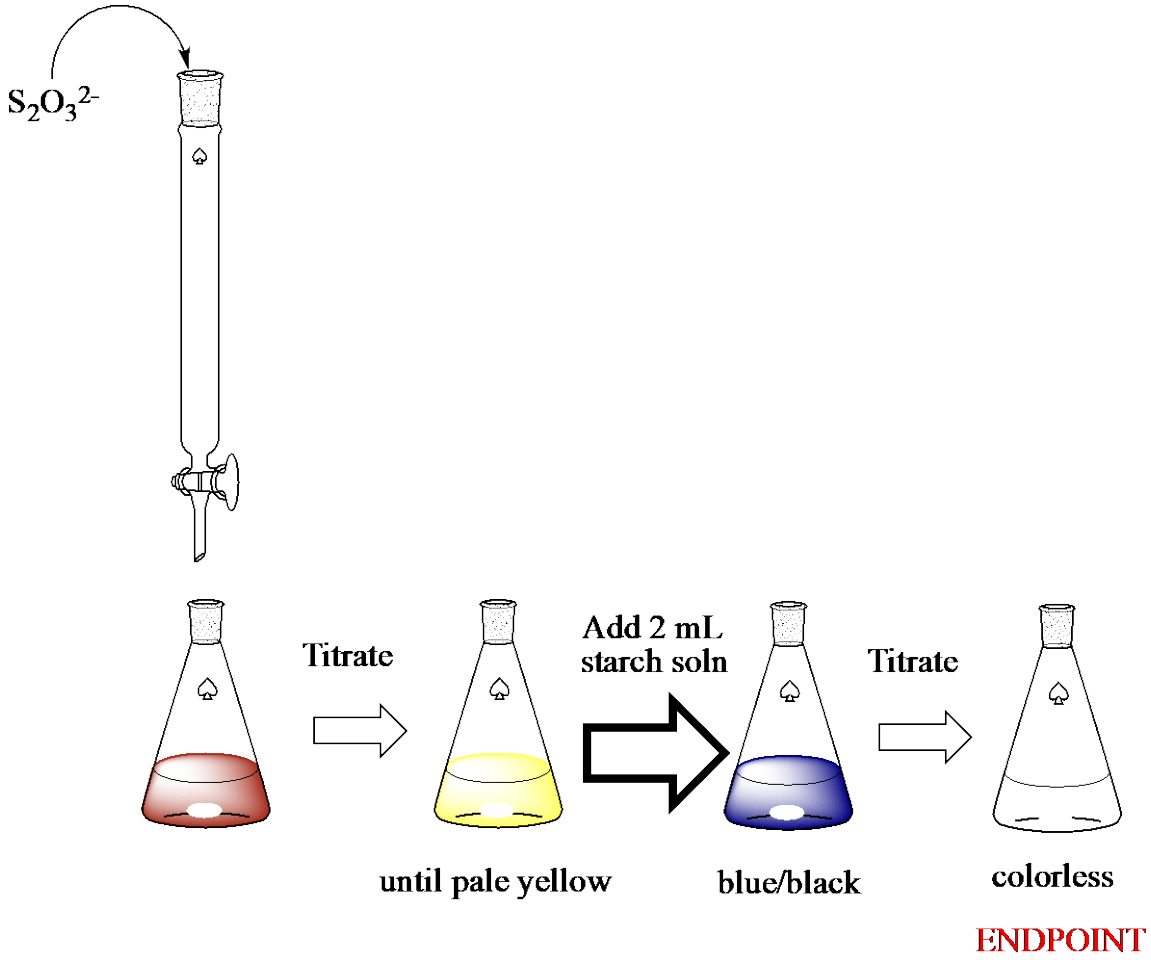
Procedure A – Standardization of $S_2O_3^{2-}$ Solution

Procedure B – Analysis of Vitamin C Tablet

Grind tablet (mortar and pestle)
Rinse into beaker with minimal DI water

# Procedure for Part A (Do this twice):




Titration:

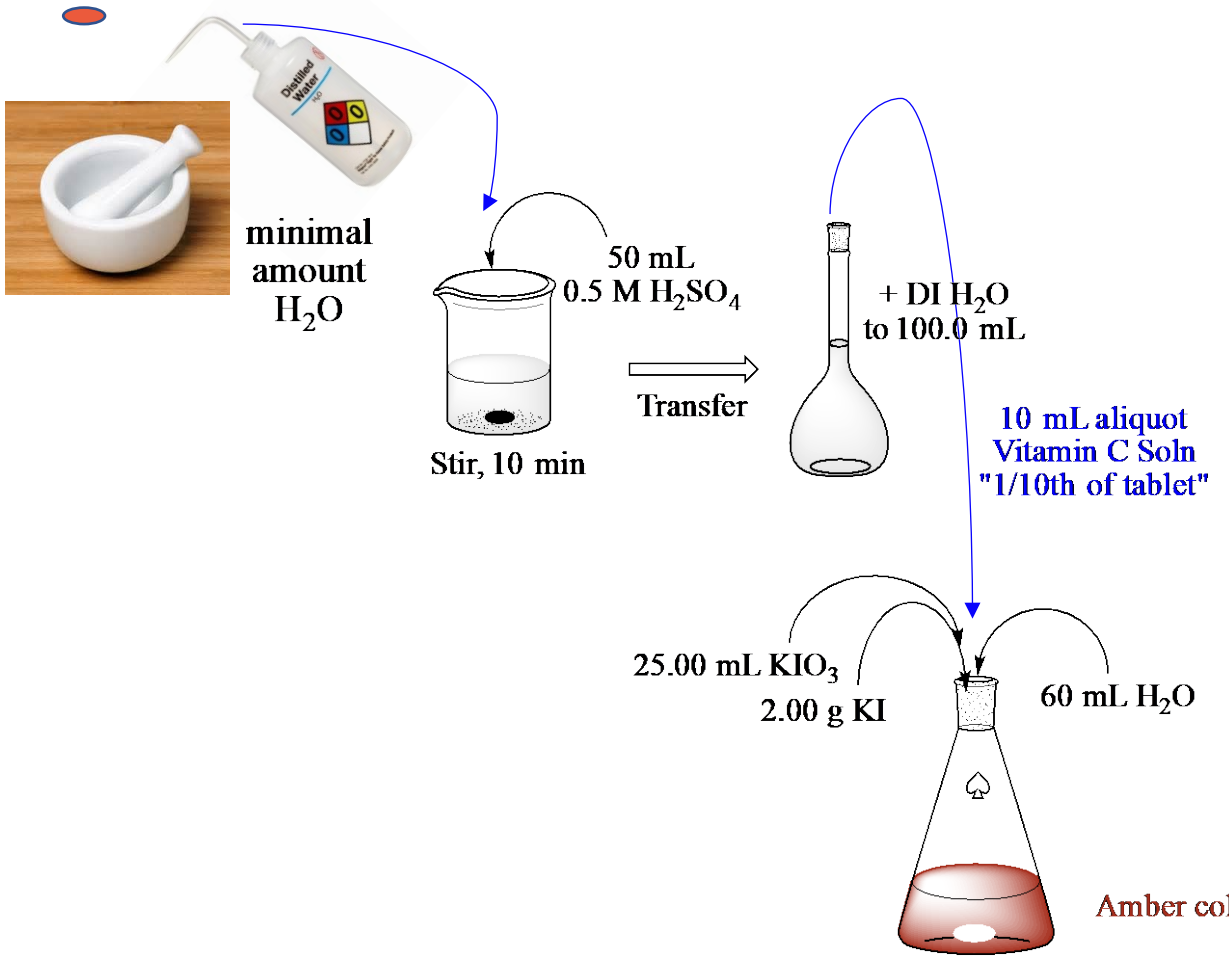
$$I_3^- + 2\,S_2O_3^{2-} \rightarrow 3\,I^- + S_4O_6^{2-}$$

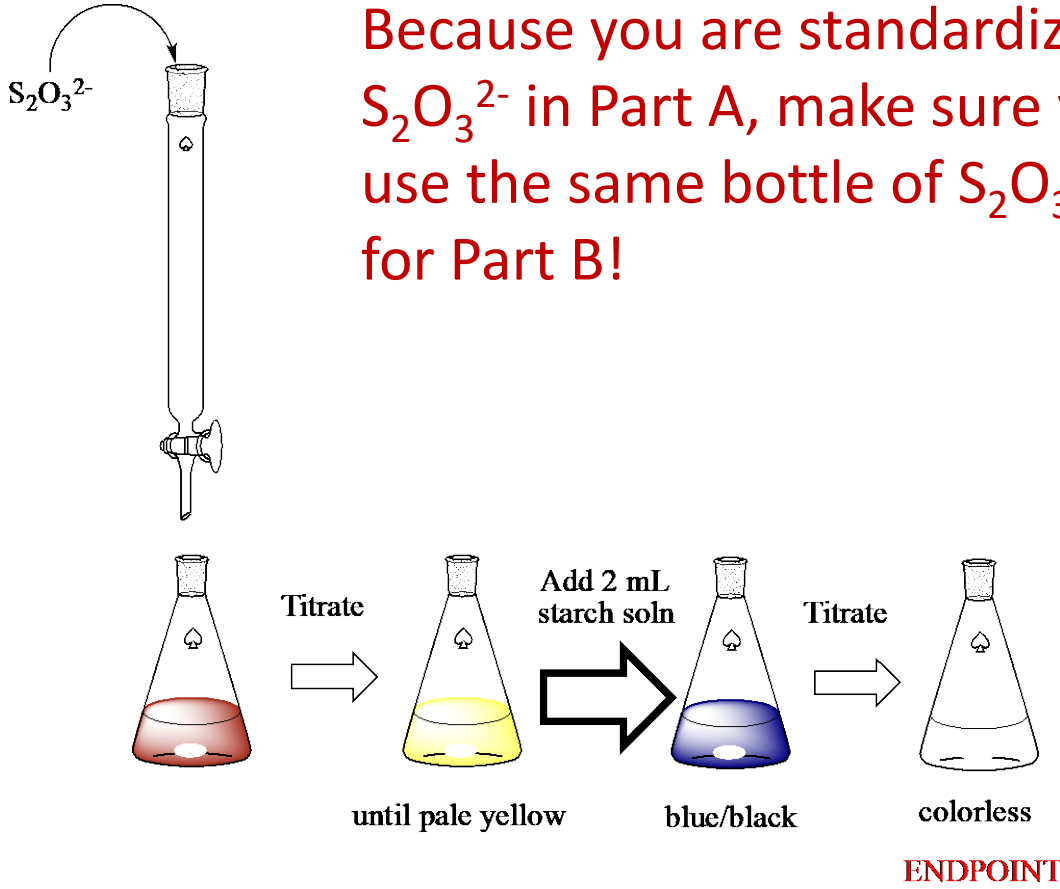
Colorless

Use Equation 5 to calculate $[S_2O_3^{2-}]$

2 runs → Average

# Procedure for Part B (Do this twice):




Because you are standardizing $S_2O_3^{2-}$ in Part A, make sure you use the same bottle of $S_2O_3^{2-}$ for Part B!

Use Eqs 1-4 → mg Vitamin C/tablet

2 runs → Average

# Safety and Clean-up

- Dispose of all waste into the appropriate waste containers
- Notebook uploads are due Sunday by 11:59PM